

PERFORMANCE TESTING

Field	Details
Date	15 March 2026
Team ID	6ab4cbbb0fc666a751b50cda
Project Name	AI-Augmented Backend Development AI FitTrack API
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. The following approach is prepared for the AI FitTrack API backend. Actual measurements must be recorded after executing the tests.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API testing). Apache JMeter is also suitable for load testing; no executed performance-test report was supplied.
Type of Testing	Load Testing / API Performance Testing
Target Module / API	Authentication APIs, Workout CRUD/Search APIs, and AI recommendation / fitness-insight APIs.
Test Environment	Local development environment; exact machine/server details were not provided in the performance-testing source.
Test Date	Not provided / to be filled after execution

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Authentication API load: repeated login/register requests.	10	60	Successful responses with acceptable response time and no significant errors.
2	Workout CRUD/search load: create, read, update, delete and search requests.	25	60	API remains responsive and data operations complete correctly.
3	AI recommendation/insight load: repeated requests to Gemini-backed endpoints.	10	60	AI requests complete successfully without excessive API/server errors.
4	Combined API load: authentication + workout + AI endpoints.	50	120	System remains stable under combined expected load.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)
1	Response Time (Avg)	< 2 seconds	Not measured	Pending execution
2	Response Time (Max)	< 5 seconds	Not measured	Pending execution
3	Throughput (Req/sec)	Record measured value	Not measured	Pending execution
4	Error Rate	< 1%	Not measured	Pending execution
5	CPU Utilization	< 80%	Not measured	Pending execution
6	Memory Utilization	< 80%	Not measured	Pending execution

Step 4: Observations & Analysis

The supplied project documentation describes a Node.js/Express backend with MongoDB/Mongoose, JWT authentication, workout APIs, and Gemini AI services. Performance values cannot be claimed without running the test scenarios. After execution, compare average/max response time, throughput, error rate, CPU and memory utilization against the stated targets. ■cite■turn0search0■

Step 5: Screenshots / Evidence

Attach screenshots from Postman, JMeter, or another performance-testing tool showing request counts, response times, throughput and error results. No performance-test screenshots were included with the supplied template, so this section remains ready for evidence.